

**NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA**  
**THEORY EXAMINATION**  
**August 01, 2026**

Roll No. 12511044

Programme : B. Tech.  
 Subject : Engineering Mechanics  
 Subject code : RAPC 101  
 Total pages : 02

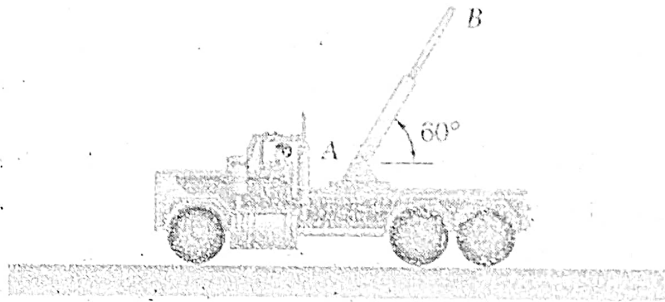
Semester : 2<sup>nd</sup>  
 Max. Time : 3 hrs.  
 Max. Marks. : 50

**Important Instructions:**

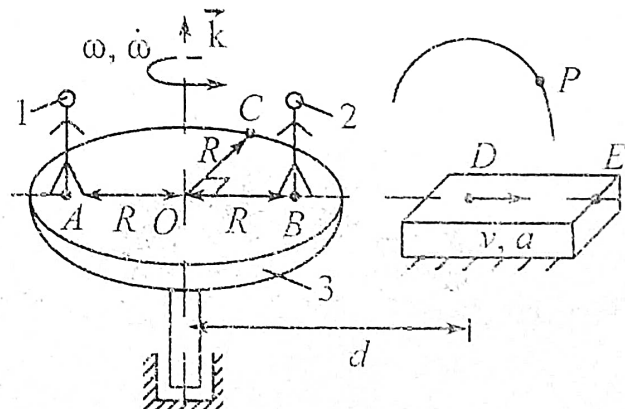
- The candidates, before starting to write the answers, should please check the Question Paper for any discrepancy and also ensure that they have been delivered the question paper of right course number and right subject title.
- Symbols and notations carry their usual meanings in the context of engineering mechanics
- Clearly state assumptions wherever taken. Assume suitable data wherever required.
- Attempt all questions

**Q1A.** A particle moves along a logarithmic spiral in the  $x - y$  plane, given by the equation  $r = ce^{\phi}$ , where  $c$  is a constant. If  $\dot{\phi} = \frac{b}{r^2}$ , where,  $b > 0$  and is a constant, find the velocity and the acceleration of the particle when the coordinates of its location are  $(r, \theta)$ . Also, find the radius of curvature ( $\rho$ ) of its trajectory at this location. Note the radius of curvature  $\rho$ , may be obtained as  $\rho = \frac{|\mathbf{v}_P|^3}{|\mathbf{v}_P \times \mathbf{a}_P|}$  [5]

**Q1B.** Consider a cylindrical-polar coordinate system with its origin at point A of a truck which is stationary w.r.t. ground. The end Point B of a boom is originally 5 m from the Point A, when the driver starts to retract the boom with constant  $\ddot{r} = -1.0 \text{ m/s}^2$  and turn it with a constant  $\ddot{\theta} = -0.5 \text{ rad/s}^2$ . At  $t = 2 \text{ s}$ , determine (a) the velocity of Point B, (b) the acceleration of Point B, (c) the radius of curvature of the trajectory of point B. [5]



**Q2A.** Two persons 1 and 2, strapped to a platform at opposite positions along AB (the points A and B are fixed to the platform), observe the motion of projectile P and an ant D moving w.r.t. ground, along OE at speed  $v \text{ m/s}$ , which is increasing at the rate  $a \text{ m/s}^2$ . The platform itself rotates relative to the ground frame with its angular velocity being  $\omega \hat{k} \text{ rad/s}$  and its angular acceleration being  $\dot{\omega} \hat{k} \text{ rad/s}^2$  relative to the ground frame. C is another point that has been fixed on the platform. [5]



- Do the two persons observe the same velocity and acceleration of the projectile P? Justify your answer.
- Find the velocity and acceleration of ant D observed with respect to the platform.
- Find the velocity of point B relative to point A with respect to (i) the platform frame and (ii) the ground frame.
- Find the velocity and acceleration of point C as seen by the first person.

- Q2B.** A wheel rolls without slipping on a fixed cylinder (**Fig. Q2B**). Knowing that at the instant shown the angular velocity of the wheel is  $10 \text{ rad/s}$  clockwise and its angular acceleration is  $30 \text{ rad/s}^2$  counterclockwise, determine the acceleration of (a) Point A, (b) Point B (c) Point C [5]
- Q3A.** A  $260 \text{ N}$  force is applied at A to the rolled-steel section as shown in the **Fig. Q3A**. Replace that force with an equivalent force-couple system at the centre C of the section. [5]

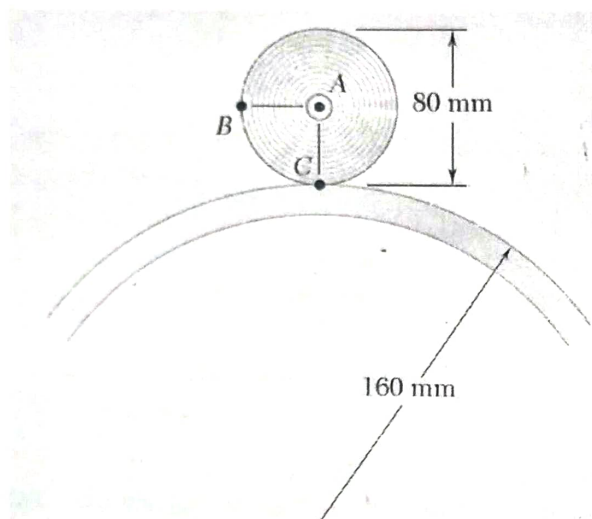


Fig. Q2B

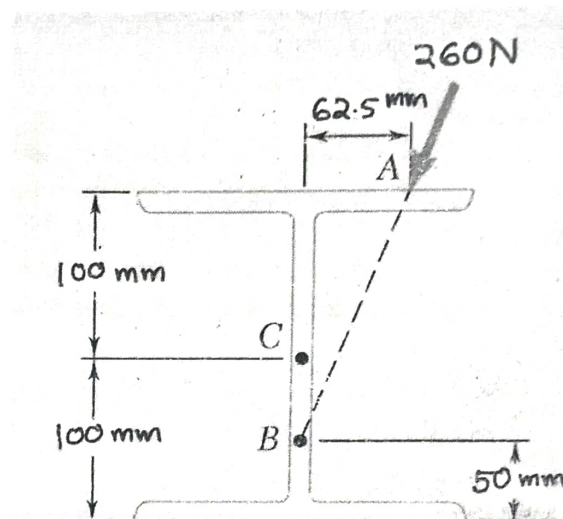


Fig. Q3A

- Q3B.** Find the inertia matrix  $\begin{bmatrix} I_x & I_{xy} \\ I_{xy} & I_y \end{bmatrix}$  of a solid circular cylinder of uniform density  $\rho$ , mass  $m$  radius  $R$  and length  $L$ . Here C denotes the centre of mass. [5]
- Q4.** There is an L shaped rod shown in **Fig. Q4** which is supported at A through a joint such that the rod is free to rotate along the vertical axis. For a given applied torque  $T$  and  $\underline{\omega}_{m/l} = \omega \hat{e}$ ,  $\left. \frac{d\hat{e}}{dt} \right|_l = 0$ . Assuming both the sections of the rod having the same mass  $m$ . Find the reaction force system at A and  $\frac{d\omega}{dt}$ . [10]

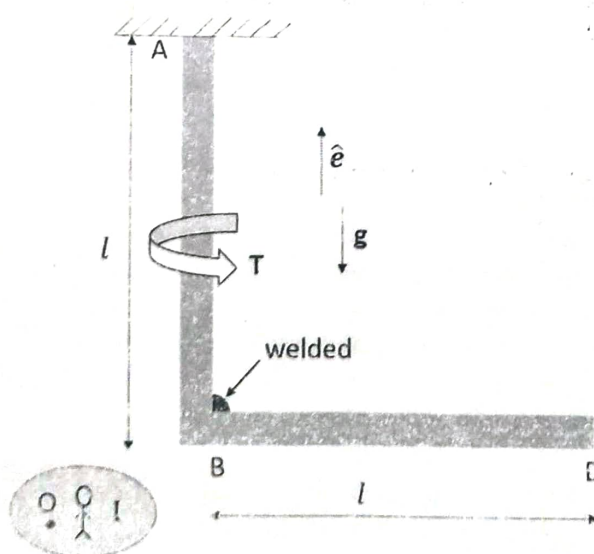


Fig. Q4

- Q5.** Prove the following expressions, the notations carry their usual meanings in the context of engineering mechanics [10]
- (a)  $\left. \frac{d\mathbf{A}}{dt} \right|_F = \left. \frac{d\mathbf{A}}{dt} \right|_m + \underline{\omega}_{m/F} \times \mathbf{A}$
- (b)  $\mathbf{V}_{P/F} = \mathbf{V}_{A/F} + \underline{\omega}_{m/F} \times \mathbf{r}_{PA} + \mathbf{V}_{P/m}$